Quiz App - JavaScript Code

```
// ---------- Question bank (multiple categories) ----------
const quizData = {
  "Web Development": [
    { q: "Which tag is used to create a hyperlink in HTML?", o: ["<link>", "<a>", "<href>", "<url>"], a:
        1 },
    { q: "Which CSS property changes the text colour?", o: ["font-color", "text-style", "color",
        "background"], a: 2 },
    { q: "Which method selects an element by its id in JavaScript?", o: ["getElementById()",
        "queryAll()", "findId()", "selectId()"], a: 0 },
    { q: "Which CSS layout model arranges items in rows and columns?", o: ["Float", "Grid", "Inline",
        "Table-cell"], a: 1 },
    { q: "What does CSS stand for?", o: ["Creative Style Sheets", "Computer Style Sheets", "Cascading
        Style Sheets", "Colorful Style Sheets"], a: 2 }
  ],
  "Python": [
    { q: "Which keyword defines a function in Python?", o: ["func", "def", "function", "define"], a: 1 },
    { q: "Which data type is immutable in Python?", o: ["List", "Dictionary", "Set", "Tuple"], a: 3 },
    { q: "What is the output of len('quiz')?", o: ["3", "4", "5", "Error"], a: 1 },
    { q: "Which symbol starts a single-line comment?", o: ["//", "#", "--", "/*"], a: 1 },
    { q: "Which library is mainly used for data analysis?", o: ["Pandas", "Pygame", "Flask", "Tkinter"],
        a: 0 }
  ],
  "Artificial Intelligence": [
    { q: "What does AI stand for?", o: ["Automated Input", "Artificial Intelligence", "Advanced
        Internet", "Applied Interface"], a: 1 },
    { q: "Which of these is a type of machine learning?", o: ["Supervised learning", "Static learning",
        "Manual learning", "Binary learning"], a: 0 },
    { q: "Which field helps computers understand human language?", o: ["Robotics", "NLP", "Networking",
        "Compilers"], a: 1 },
    { q: "Neural networks are inspired by the ...", o: ["Human brain", "Solar system", "Database",
        "Keyboard"], a: 0 },
    { q: "Which is a popular Python library for deep learning?", o: ["Bootstrap", "TensorFlow", "jQuery",
         "Django"], a: 1 }
  ],
  "General Knowledge": [
    { q: "What is the capital of India?", o: ["Mumbai", "Chennai", "New Delhi", "Kolkata"], a: 2 },
    { q: "Which planet is known as the Red Planet?", o: ["Venus", "Mars", "Jupiter", "Saturn"], a: 1 },
    { q: "How many days are there in a leap year?", o: ["365", "364", "366", "367"], a: 2 },
    { q: "Which is the largest ocean on Earth?", o: ["Atlantic", "Indian", "Arctic", "Pacific"], a: 3 },
    { q: "Who wrote the Indian national anthem?", o: ["Rabindranath Tagore", "Bankim Chandra", "Gandhi",
        "Nehru"], a: 0 }
  ]
};

const TIME_LIMIT = 60; // seconds per quiz

// ---------- State ----------
let category = "", questions = [], current = 0, selected = null, answers = [], timeLeft = 0, timerId =
    null;
const $ = id => document.getElementById(id);

// Escape text so symbols like <a> are shown as text, not treated as HTML
const esc = t => String(t).replace(/&/g, "&amp;").replace(/</g, "&lt;").replace(/>/g, "&gt;");

function showScreen(id) {
  document.querySelectorAll(".screen").forEach(s => s.classList.remove("active"));
  $(id).classList.add("active");
}

// ---------- Categories + best score (Local Storage) ----------
function getBest(cat) { return localStorage.getItem("best_" + cat); }

function renderCategories() {
  $("categories").innerHTML = Object.keys(quizData).map(cat => {
    const best = getBest(cat);
    return `<div class="cat" onclick="startQuiz('${cat}')">
      <h3>${cat}</h3><small>${quizData[cat].length} questions</small>
      <small>Best score: ${best !== null ? best + "%" : "--"}</small></div>`;
  }).join("");
}

// ---------- Randomize (Fisher-Yates shuffle) ----------
function shuffle(arr) {
  const a = [...arr];
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  for (let i = a.length - 1; i > 0; i--) {
    const j = Math.floor(Math.random() * (i + 1));
    [a[i], a[j]] = [a[j], a[i]];
  }
  return a;
}

// ---------- Start quiz ----------
function startQuiz(cat) {
  category = cat;
  questions = shuffle(quizData[cat]);
  current = 0; answers = []; selected = null;
  timeLeft = TIME_LIMIT;
  $("catName").textContent = cat;
  showScreen("quizScreen");
  showQuestion();
  updateTimer();
  clearInterval(timerId);
  timerId = setInterval(() => {
    timeLeft--;
    updateTimer();
    if (timeLeft <= 0) finishQuiz(true);
  }, 1000);
}

function updateTimer() {
  const m = String(Math.floor(timeLeft / 60)).padStart(2, "0");
  const s = String(timeLeft % 60).padStart(2, "0");
  $("timer").textContent = `${m}:${s}`;
  $("timer").classList.toggle("warn", timeLeft <= 10);
}

// ---------- Display question dynamically ----------
function showQuestion() {
  const q = questions[current];
  selected = null;
  $("qCount").textContent = `Question ${current + 1} of ${questions.length}`;
  $("progressBar").style.width = (current / questions.length * 100) + "%";
  $("question").textContent = q.q;
  $("options").innerHTML = q.o.map((opt, i) =>
    `<button class="option" onclick="selectOption(${i}, this)">${esc(opt)}</button>`).join("");
  $("nextBtn").disabled = true;
  $("nextBtn").textContent = current === questions.length - 1 ? "Submit" : "Next";
}

function selectOption(i, el) {
  selected = i;
  document.querySelectorAll(".option").forEach(o => o.classList.remove("selected"));
  el.classList.add("selected");
  $("nextBtn").disabled = false;
}

$("nextBtn").onclick = () => {
  answers[current] = selected;
  if (current === questions.length - 1) return finishQuiz(false);
  current++;
  showQuestion();
};

// ---------- Finish: score, summary, review, best score ----------
function finishQuiz(timeUp) {
  clearInterval(timerId);
  const total = questions.length;
  const attempted = answers.filter(a => a !== undefined && a !== null).length;
  const correct = questions.filter((q, i) => answers[i] === q.a).length;
  const percent = Math.round(correct / total * 100);

  $("banner").textContent = timeUp ? "Time's up! Your quiz was submitted automatically." : "";
  $("percent").textContent = percent + "%";
  $("scoreText").textContent = `${correct} / ${total}`;
  $("sTotal").textContent = total;
  $("sAttempted").textContent = attempted;
  $("sCorrect").textContent = correct;
  $("sWrong").textContent = attempted - correct;

  const prev = getBest(category);
  if (prev === null || percent > +prev) localStorage.setItem("best_" + category, percent);
  $("bestScore").textContent = "Best score in " + category + ": " + getBest(category) + "%";
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  $("review").innerHTML = questions.map((q, i) => {
    const ok = answers[i] === q.a;
    const yours = answers[i] === undefined || answers[i] === null ? "Not answered" : q.o[answers[i]];
    return `<div class="rev ${ok ? "right" : "wrong"}"><h4>Q${i + 1}. ${esc(q.q)}</h4>
      Your answer: ${esc(yours)}<br>Correct answer: ${esc(q.o[q.a])}</div>`;
  }).join("");
  showScreen("resultScreen");
}

// ---------- Restart / another category ----------
$("restartBtn").onclick = () => startQuiz(category);
$("anotherBtn").onclick = () => { renderCategories(); showScreen("startScreen"); };

renderCategories();
```